

Guilherme Ruiz Ferreira, PhD

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Work History

- 2021 – 2025  **PhD in Particle Physics**, University of Manchester — Manchester, UK
Developed and tested detector instrumentation, carried out experimental measurements and simulations, built and maintained analysis pipelines, conducted sensitivity studies, and presented and published results with international collaborators.
- 2020 – 2021  **Network Control Manager**, Amazon Logistics — London, UK
Modelled warehouse operations, analysed data to resolve connectivity issues, automated workflows and reporting (Python, SQL, VBA), built internal web tools (HTML/CSS), and delivered actionable insights to stakeholders in a fast-paced environment.
- 2014 – 2016  **Undergraduate Researcher**, University of São Paulo (USP) — São Paulo, Brazil
Contributed to detector construction and testing for a major collider experiment upgrade, participated in collaborative data taking and analysis, and presented research at an international symposium.

Education

- 2018 – 2021  **MSc in Physics** — Grade: Distinction
University College London (UCL) — London, UK
- 2013 – 2018  **BSc in Physics with Astronomy** — Grade: 7/10
University of São Paulo (USP) — São Paulo, Brazil

Skills

- Languages  Native Portuguese — Fluent English — Basic Spanish.
- Systems  Windows, macOS, Linux, Docker, Microsoft Office (VBA), COMSOL.
- Programming  Python, Swift, Java, ROOT, C/C++, Mathematica, $\text{\LaTeX}$, HTML/CSS, SQL, Git, InfluxDB.
- Electronics  Arduino, microcontrollers, nuclear instrumentation modules, data acquisition systems.
- Academic  Data analysis, research, scientific writing, teaching/tutoring.
- Aviation  Private Pilot License (PPL), ~ 100 flight hours.

Research Publications

- 1 'Cosmic Ray Measurements Using Charge and Light Readout in a Pixelated Liquid Argon Time Projection Chamber'.  DOI: [10.48550/arXiv.2512.10830](https://doi.org/10.48550/arXiv.2512.10830).
- 2 'Opportunities and challenges to study solar neutrinos with a Q-Pix pixel readout'.  DOI: [10.1103/xq7l-lv7j](https://doi.org/10.1103/xq7l-lv7j).
- 3 'First demonstration of a combined light and charge pixel readout on the anode plane of a LArTPC'.  DOI: [10.1088/1748-0221/19/11/p11010](https://doi.org/10.1088/1748-0221/19/11/p11010).
- 4 'Distinguishing circumstellar from stellar photometric variability in Eta Carinae'.  DOI: [10.1093/mnras/stz067](https://doi.org/10.1093/mnras/stz067).